BONUS OP

Chemical Engineering Plebe Majors Fair

21 OCT 2025 from ~1220 to ~1350
Thayer Hall Room 336, 368, or 370[1]

30 minutes = 5 points[2]
Max 1.5 hours (15 points)

Notes:
1. We will only occupy one room.
2. Sign in and out on the provided roster with time in and time out. Interact with prospective cadets and answer questions. Stay active. Try not to congregate in “friends” clusters.

CH365 Chemical Engineering Thermodynamics

Lesson 23
Calculation of Entropy and
Entropy Changes for an Ideal Gas

Summary of Section 5.4

There exists a property called entropy S, which is an intrinsic property of a system, functionally related to the measurable coordinates characterizing the system. For a reversible process, changes in this property are given by Eq. 5.1.

$$dS^t = \frac{dQ_{rev}}{T} \qquad \text{Eq. 5.1} \qquad \text{(Axiom to 2nd Law)}$$

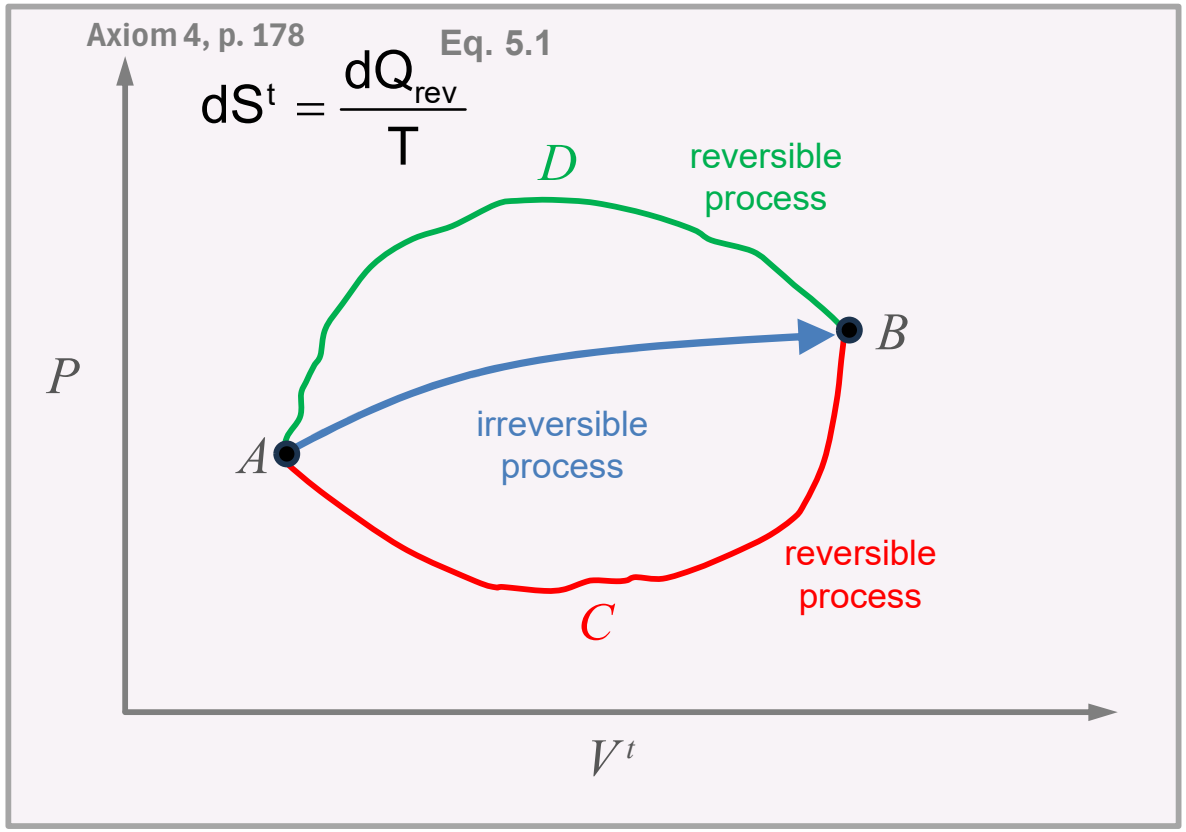
The change in entropy for any system undergoing a finite reversible process is:

$$\Delta S^t = \int \frac{dQ_{rev}}{T} \qquad \text{Integral of Eq. 5.1}$$

When a system undergoes an irreversible process between two equilibrium states, the irreversible path cannot be directly integrated. The entropy change of the system is evaluated by integrating Eq. 5.1 along an arbitrarily chosen reversible process that accomplishes the same change of state as the actual process. Because entropy is a state function, the entropy changes of the irreversible and reversible processes are identical.

Entropy - State Function

I. Changes are Independent of Path



$$\Delta S^t_{ACB} = \int_{ACB} \frac{dQ_{rev,ACB}}{T} \quad \Delta S^t_{ADB} = \int_{ADB} \frac{dQ_{rev,ADB}}{T} \quad \Delta S^t_{AB} = \int_{AB} \frac{dQ_{irrev,AB}}{T}$$

The entropy change is the same for irreversible and reversible processes.

$$\Delta S^t_{AB} = \Delta S^t_{ACB} = \Delta S^t_{ADB}$$

Entropy - State Function

II. Derives from a Combination of Other State Functions

$$dU = dQ + dW$$ Eq. 2.4

$$dU = dQ_{rev} - PdV$$ First Law, Eq. 2.7, reversible

Eq. 1.3 dW=-PdV

$$H = U + PV$$ Definition of enthalpy, Eq. 2.10

$$dH = dU + PdV + VdP$$ Differentiate

$$dH = dQ_{rev} - \cancel{PdV} + \cancel{PdV} + VdP = dQ_{rev} + VdP$$

$$dQ_{rev} = \underbrace{dH} - \underbrace{VdP} \qquad \underbrace{dH} = C_P^{ig}dT \quad \text{and} \quad \underbrace{V} = \frac{R \cdot T}{P}$$ Ideal Gas

$$dQ_{rev} = C_P^{ig} \cdot dT - R \cdot T \cdot \frac{dP}{P}$$

$$\underbrace{\frac{dQ_{rev}}{T}} = C_P^{ig}\frac{dT}{T} - R \cdot \frac{dP}{P} \qquad dS = \underbrace{\frac{dQ_{rev}}{T}}$$ Eq. 5.1

$$dS = C_P^{ig}\frac{dT}{T} - R \cdot \frac{dP}{P}$$

$$\frac{dS}{R} = \frac{C_P^{ig}}{R}\frac{dT}{T} - d\ln P \quad \Longleftrightarrow \quad \frac{\Delta S}{R} = \int_{T_0}^{T} \frac{C_P^{ig}}{R}\frac{dT}{T} - \ln\frac{P}{P_0}$$ Eq. 5.10

Integrated Forms of Entropy (Eq. 5.10)

$$\mathrm{ICPS} = \int_{T_0}^{T} \frac{C_P^{ig}}{R}\frac{dT}{T} = A \cdot \ln\frac{T}{T_0} + \left[B + \left(C + \frac{D}{T_0^2\,T^2}\right)\cdot\left(\frac{T+T_0}{2}\right)\right]\cdot(T - T_0)$$

Eq. 5.11

$$\mathrm{MCPS} = \frac{\left\langle C_P^{ig}\right\rangle_S}{R} = A + \left[B + \left(C + \frac{D}{T_0^2\,T^2}\right)\cdot\left(\frac{T+T_0}{2}\right)\right]\cdot\left(\frac{T - T_0}{\ln(T/T_0)}\right)$$

Eq. 5.13

where $$\left\langle C_P^{ig}\right\rangle_S = \frac{\int_{T_0}^{T} C_P^{ig} dT/T}{\ln(T/T_0)}$$

Eq. 5.12

$$\frac{\Delta S}{R} = \int_{T_0}^{T} \frac{C_P^{ig}}{R}\frac{dT}{T} - \ln\frac{P}{P_0}$$

Eq. 5.10

$$\frac{\Delta S}{R} = \mathrm{ICPS} - \ln\frac{P}{P_0}$$

$$\frac{\Delta S}{R} = \mathrm{MCPS}\cdot\ln\left(\frac{T}{T_0}\right) - \ln\frac{P}{P_0}$$

Eq. 5.14

Example 5.4

Methane gas at 550 K and 5 bar undergoes a reversible adiabatic expansion to 1 bar. Assuming methane to be an ideal gas at these conditions, find its final temperature.

$$\Delta S = 0$$

$$\frac{\Delta S}{R} = MCPS \cdot \ln\left(\frac{T}{T_0}\right) - \ln\frac{P}{P_0} \qquad \text{Eq. 5.14}$$

$$MCPS = A + \left[B + \left(C + \frac{D}{T_0^2\, T^2}\right) \cdot \left(\frac{T + T_0}{2}\right)\right] \cdot \left(\frac{T - T_0}{\ln(T / T_0)}\right) \qquad \text{Eq. 5.13}$$

T_0, P_0, and P are all known

T is the only unknown

Solve in Mathematica

Example 5.4.nb - Wolfram Mathematica

File Edit Insert Format Cell Graphics Evaluation Palettes Window Help

Insert Cell...

```
R = 8.314; (*J/mol*K*)
a = 1.702;
b = 9.081 * 10^-3;
c = -2.164 * 10^-6;
d = 0 * 10^5;

T0 = 550; (*K*)
P0 = 5; (*bar*)
P = 1; (*bar*)

MCPS = a + (b + (c + d/(T0^2 * T^2)) * ((T + T0)/2)) * ((T - T0)/Log[T / T0]);
```

In[24]:=
```
eq1 = 0 == MCPS * Log[T / T0] - Log[P/P0];
```

In[23]:=
```
Solve[eq1, T, Reals] // Quiet
```

Out[23]=
```
{{T → 411.3344399}, {T → 8567.069029}}
```

ANS, K

100%

Questions